English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Stream 2, 20 June 2026 Examination problems _____ *The exact grading formula depends on the results. As an approximation, use 2 + the number of points. Time allowed: 3 hours. Good luck! You may use*

$$\int_0^1 p^k(1-p)^{n-k}\,\mathrm{d}p = \frac{k!(n-k)!}{(n+1)!}.$$

Problem 1. (1.5 points) The following questions are separate.

1. (0.25 points) Roll two fair dice. Let A mean the first result is even, B the second is even, and C the sum is even. Prove that A, B, C are pairwise but not jointly independent. Are A, B conditionally independent given C?

2. (0.5 points) A coin's probability p of heads is chosen uniformly on $[0, 1]$. Toss it n times and let S_n count heads. Prove that S_n is uniform on $\{0, 1, \ldots, n\}$. Then find the probability of heads on toss $n + 1$ given k observed heads.

3. (0.25 points) A flea jumps between the vertices of an equilateral triangle, each time choosing either other vertex equally likely, independently of the past. Find the probability p_n of being at its starting vertex after n jumps, and $\lim_{n\to\infty} p_n$.

4. (0.5 points) Let $X_i \overset{iid}{\sim} \mathrm{Exp}(\lambda)$. Calculate $\mathbb{E}[X_1 \mid X_1 + X_2 = t]$ and $\mathrm{Cor}(X_1, X_1 + X_2)$.

Problem 2. (1.25 points) An insect lays $N \sim \mathrm{Poi}(\lambda)$ eggs. Each hatches independently with probability p. Let X count hatched eggs and $Y = N - X$ unhatched eggs.

1. (0.25 points) Use generating functions to find the distribution of X.

2. (0.5 points) Prove that X, Y are independent and specify their distributions.

3. (0.5 points) Find $\mathrm{Cor}(N, X)$ and $\mathbb{E}[N \mid X = k]$.

Problem 3. (1 point) During an Operating Systems examination, Angel enters an N-character password in a remote terminal, pressing one key exactly every T seconds. He then presses `Enter`, registered instantaneously at $t_E = (N+1)T$. TCP with a reordering buffer preserves character order. Character i, $1 \le i \le N$, reaches the server after a random delay D_i.

1. (0.2 points) If $F(x) = P(D < x)$ is the distribution function of each packet's delay, derive the probability P_{success} that the password is registered correctly: all N characters arrive before `Enter`.

2. (0.3 points) Unsure how to solve a question, Angel writes an essay explaining why he deserves points. It has $N = 10000$ characters typed at intervals $T = 0.2$ seconds. A shared server queue processes a random number X_k of characters each second, independently and identically distributed with mean 4 and variance 2. On pressing `Enter`, all unprocessed characters are lost. Use the central limit theorem to approximate the mean number of *whole words* lost, assuming six characters per average word.

3. (0.5 points) Real packet delays are dependent. Model them by $D_i = \rho D_{i-1} + \varepsilon_i$, with $D_1 = \varepsilon_1$, memory coefficient $\rho \in (0, 1)$, and independent $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ representing white noise. The last character has the least time before `Enter`. Find:

   (a) the exact expectation $\mathbb{E}[D_N]$ and variance $\mathrm{Var}(D_N)$;

   (b) a formula in terms of Φ for $P(R_N < t_E)$, the probability that the last character arrives in time.

Problem 4. (0.75 points) A rectangular sheet has vertices A, B, C, D and $AD = BC = 10$ cm. Points F, K are chosen at random on sides AD, CD so that folding along FK brings D onto side AB. Find the expected area of triangle FKD.